

Erdős Problem #920

EP-920

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Let $f_k(n)$ be the maximum possible chromatic number of a graph with n vertices which contains no K_k .

Is it true that, for $k \geq 4$, $f_k(n) \gg \frac{n^{1-\frac{1}{k-1}}}{(\log n)^{c_k}}$ for some constant $c_k > 0$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>